Modelling and cheminformatics approaches for peptide analysis and design

Rodrigo Ochoa, Principal Data Scientist

InSilico Peptide Discovery

Agenda

01 Novo introduction and computational efforts

02 ML impact to predict peptide properties

03 PepFuNN: python package for peptide analysis

04: AI-driven approaches to predict conformers of cyclic peptides



Novo at a glance

Novo at a **glance**

Novo Nordisk is a leading global healthcare company, founded in 1923 and headquartered in Denmark.

Our purpose is to drive change to defeat serious chronic diseases, built upon our heritage in diabetes.

We do so by pioneering scientific breakthroughs, expanding access to our medicines, and working to prevent and ultimately cure disease.

1. https://companiesmarketcap.com/pharmaceuticals/largest-pharmaceutical-companies-by-market-cap/ (As of 6 August 2025).

Supplier of nearly
50%
of the world's insulin

Net sales
290
billion DKK

Affiliates in
80
countries

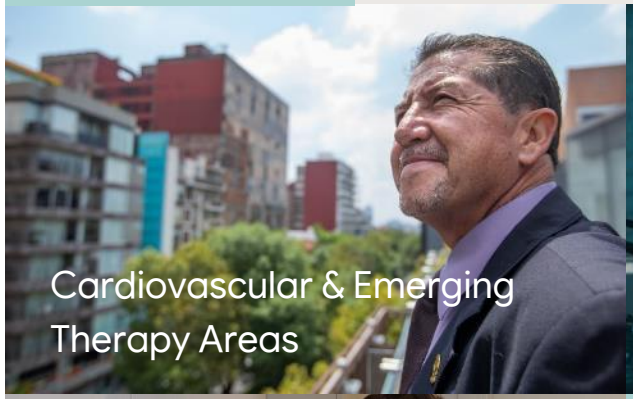
More than
77,349
employees

Investing more than
52
billion DKK in R&D

R&D facilities in China, Denmark, UK, US and India

Strategic production sites in Belgium, Brazil, China, Denmark, France, Italy and US

Globally, serving more than
45.2
million people living with serious chronic diseases







A top ten
pharma company measured by market value[1]

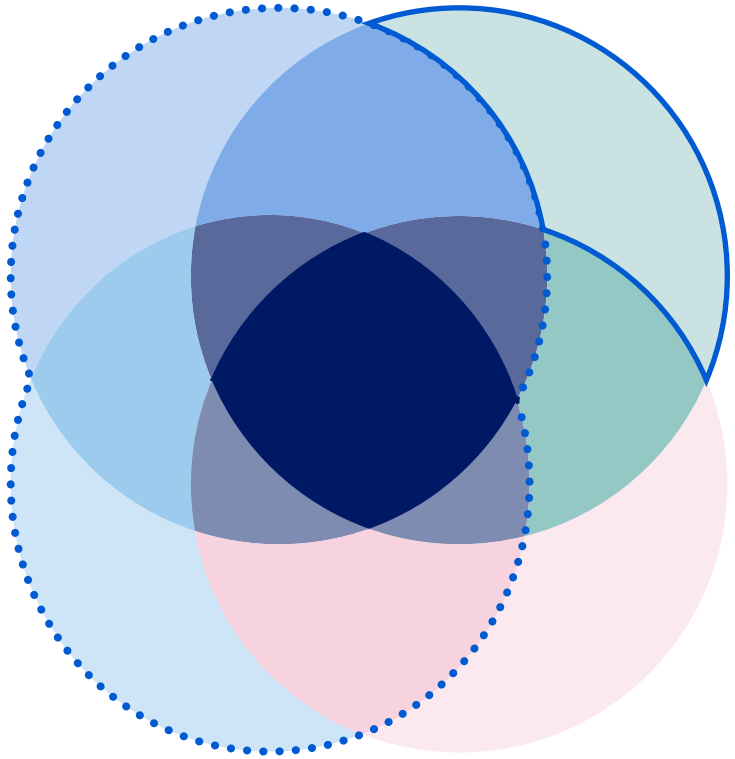


Metabolic and cardiovascular health

Metabolic and cardiovascular health is a key priority to Novo Nordisk. We stride to provide life-changing and innovative treatments for patients that deliver complementary effects across the spectrum of **metabolic and cardiovascular diseases.**

Diabetes
537m people[1]

Cardiovascular disease
>500m people[3]



Cardiometabolic diseases cover a group of interconnected chronic medical conditions affecting the heart, blood vessels, liver (MASH*), and kidneys and originates from chronic metabolic diseases such as obesity and diabetes.

Cardiovascular disease is a group of serious diseases affecting the heart or blood vessels. At Novo Nordisk, we focus on pathologies related to atherosclerosis and heart failure. Some of these conditions are caused by metabolic dysfunctions, while others arise independently of them.

Obesity
~1bn people[2]

MASH*
>250m people[4]

lillustrative

**MASH: : Metabolic dysfunction-associated steatohepatitis. MASH is also known as nonalcoholic steatohepatitis (NASH). Among people living with overweight and obesity or people living with T2D, one in three live with MASH[5,6].*

Our core technology platforms



Proteins, peptides, and antibodies

Our core expertise is in developing therapeutic proteins, peptides, and antibodies. From ideation and synthesis, to engineering and formulation.



Oligonucleotides

Oligonucleotides such as RNA interference (RNAi) therapies harness biological processes to selectively silence genes that cause or contribute to disease.



Small molecules

Small molecules can be made chemically on a large scale and conveniently taken as an oral tablet without dosing conditions.



Gene therapies

Inserting, deleting, modifying or replacing DNA in the human genome holds the promise of delivering a true cure for genetic diseases.



Cell therapy

Pluripotent stem cells, enable us to develop cells to halt or perhaps reverse progression of diseases which are caused when cell types become lost or damaged.

Mission & Purpose | Digital Chemistry & Design (DCD)

Apply state-of-the-art computational approaches, platforms & tools to accelerate the discovery and development of differentiated therapeutics with an increased probability of clinical success in order to sustain the portfolio pipeline, address unmet medical need and improve outcomes for patients worldwide.



R&D Data Science | Digital Chemistry & Design

WHAT WE DO

Deliver the Therapeutic Pipeline: Collaborate across R&ED to support the delivery of differentiated therapeutic molecules across the portfolio through applied *in silico* therapeutic design & optimization

Technology Enablement: Accelerate the discovery of differentiated therapeutics through the collaborative development, evaluation and application of innovative computational tools, platforms and capabilities

OUR IMPACT

i) Advance & deliver the pipeline of therapeutic molecules across all modalities & therapeutic areas

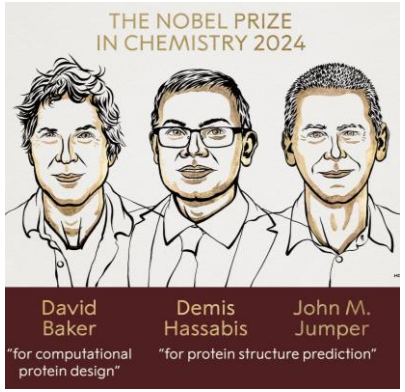
ii) Digital innovation to reduce cycle times and cost of therapeutics with an increased probability of clinical success

iii) Enterprise scalable digital platforms, models and computational tools for democratized analytics & insights

Significant Advances in AI for Drug Discovery

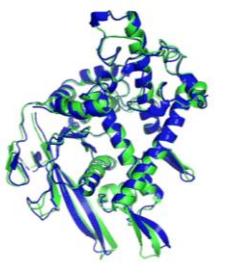
Foundational Models Shaping the Future of Healthcare & Life Sciences

Google Revolutionizes Protein Structure Prediction with AlphaFold

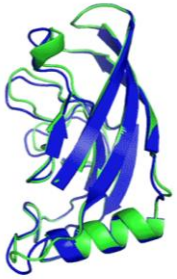


"This is the first time a serious scientific problem has been solved by AI."

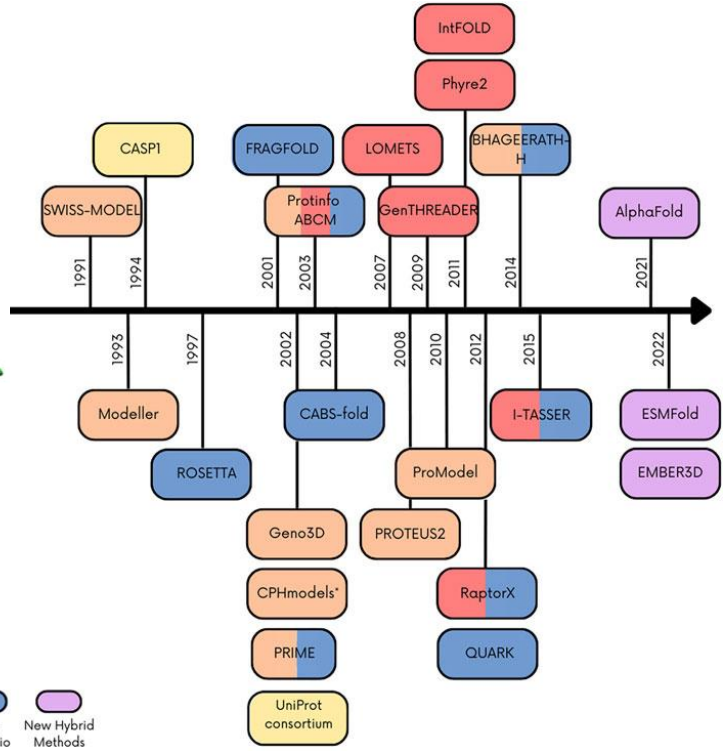
- John Moult (Critical Assessment of Protein Structure Prediction, CASP co-founder)



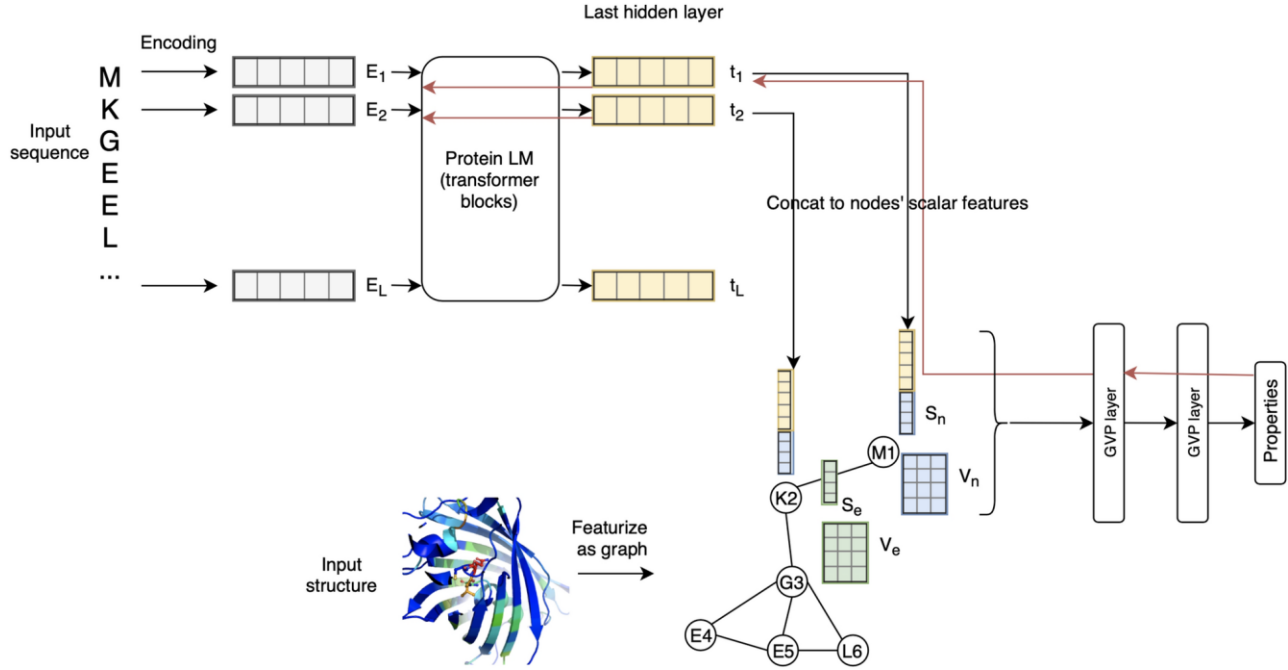
T1037 / 6vr4
90.7 GDT
(RNA polymerase domain)



T1049 / 6y4f
93.3 GDT
(adhesin tip)



Accelerating Therapeutic R&D with Large Language Models (LLMs)



LM-GVP: an extensible sequence and structure informed deep learning framework for protein property prediction

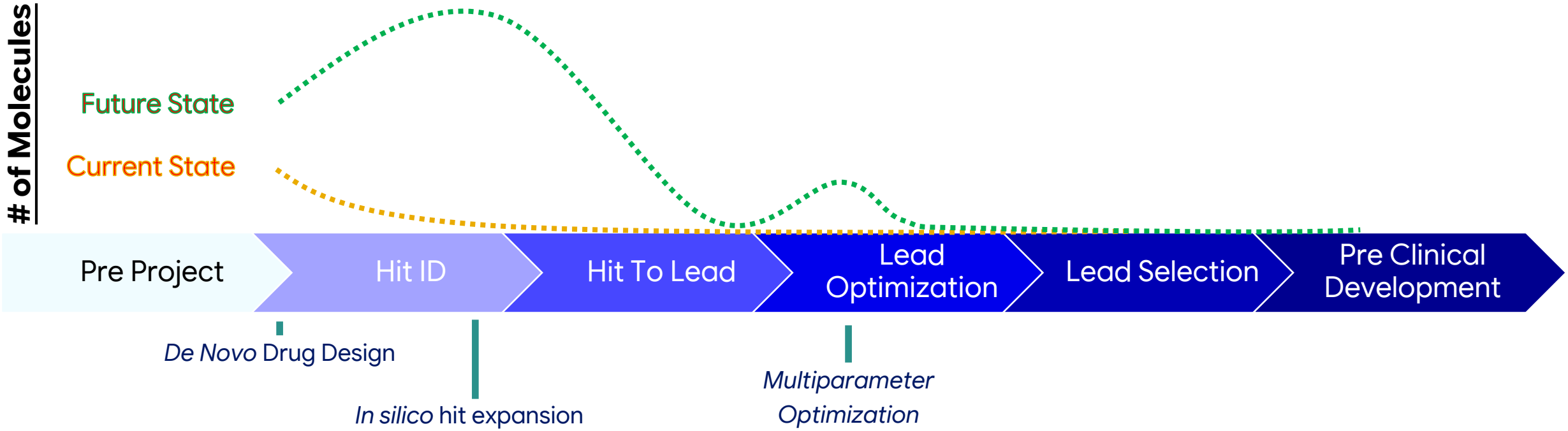
Zichen Wang, Steven A. Combs, Ryan Brand, Miguel Romero Calvo, Panpan Xu, George Price, Nataliya Golovach, Emmanuel O. Salawu, Colby J. Wise, Sri Priya Ponnapalli & Peter M. Clark

Scientific Reports **12**, Article number: 6832 (2022) | Cite this article

14k Accesses | **29** Citations | **12** Altmetric | Metrics

Fuelling R&D Transformation Through Generative AI

Impact Across the R&ED Value Chain



Constrained de novo design to increase probability of technical success, reduce cycle time & cost

Peptide analysis and design toolbox

Cheminformatics and SAR analyses

- Data visualization
- Property calculation [physchem, sequence liabilites]
- Library design

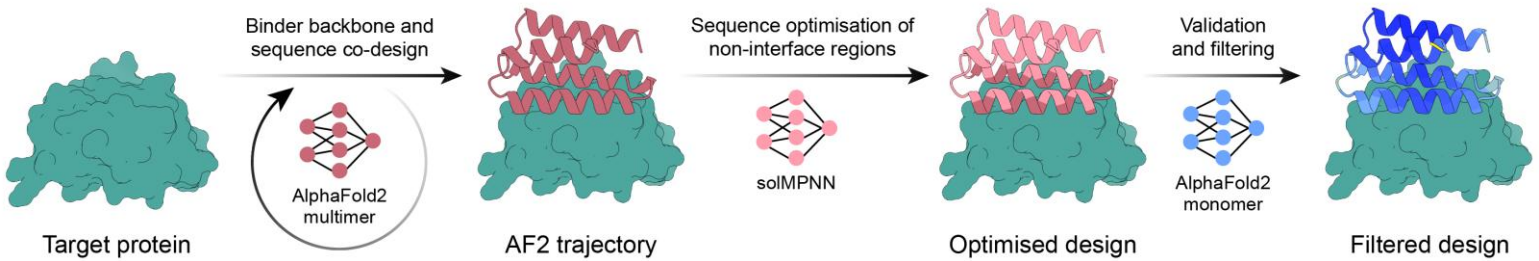
Structure-based modelling

- *De novo* design
- Target-API complex prediction [binding, dynamics]
- Modelling ncAA-containing peptides
- Formulation [effect of excipients]

AI/ML methods

- Property models
- Multi-parameter optimization

Example: Bindcraft



ML impact to predict peptide properties

- Collaboration between:







Fernández-Díaz *et al.*
Journal of Cheminformatics (2025) 17:185
https://doi.org/10.1186/s13321-025-01115-z

Journal of Cheminformatics

RESEARCH Open Access

How to build machine learning models able to extrapolate from standard to modified peptides

Raúl Fernández-Díaz[1,3,4,5*], Rodrigo Ochoa[2], Thanh Lam Hoang[1], Vanessa Lopez[1] and Denis C. Shields[3,4,5*]

Datasets

- We prepared a group of benchmark datasets that are available to test and compare any future peptide representation methods that may be proposed.
- These datasets comprised four pairs, each containing either canonical or non-canonical peptides. The datasets cover different peptide properties, namely:
  - protein-peptide binding affinity
  - cell permeability
  - antibacterial activity
  - antiviral activity

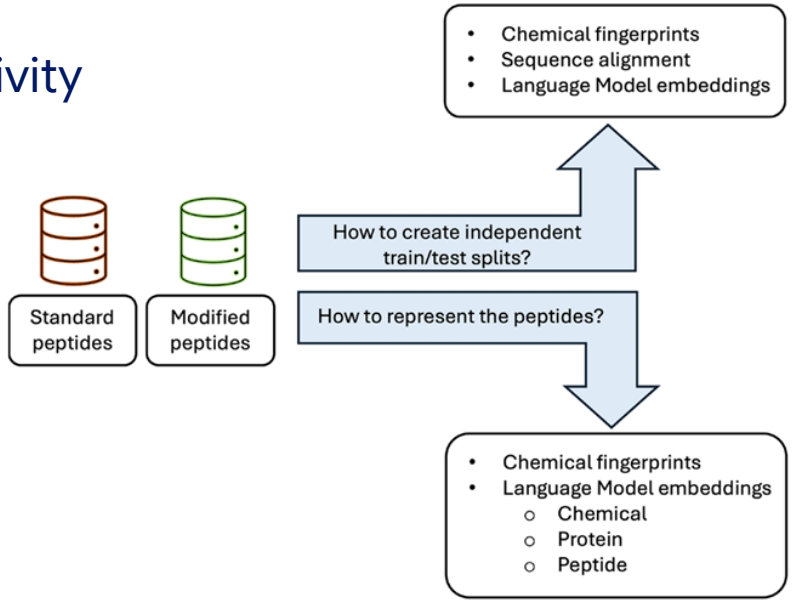


Fig. 1 Visual summary of main research questions

Table 1 Benchmark datasets. *: Dataset has been modified in this study (More details in Supplementary SA)

Dataset	Type	Task	Size	Source
Protein-peptide binding affinity	Standard	Regression	1,002	[30]
Protein-peptide binding affinity	Modified	Regression	299	[30], *
Cell penetration	Standard	Classification	2,324	[30], *
Cell penetration	Modified	Classification	480	[30], *
Antibacterial	Standard	Classification	9,855	[24], *
Antibacterial	Modified	Classification	1,880	[33], *
Antiviral	Standard	Classification	4,754	[24], *
Antiviral	Modified	Classification	444	[33], *

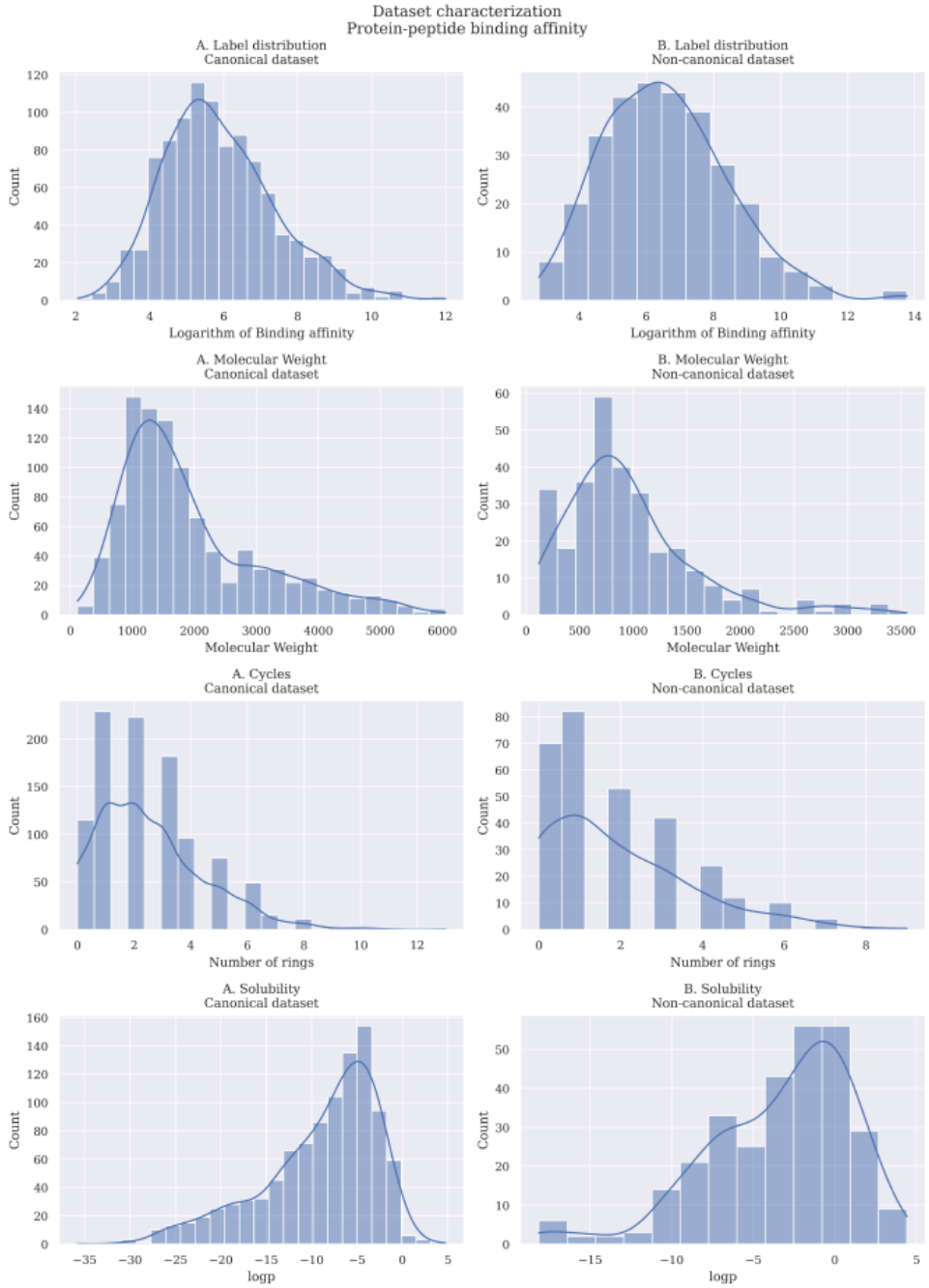


Figure 1: Dataset characterization for the binding datasets.

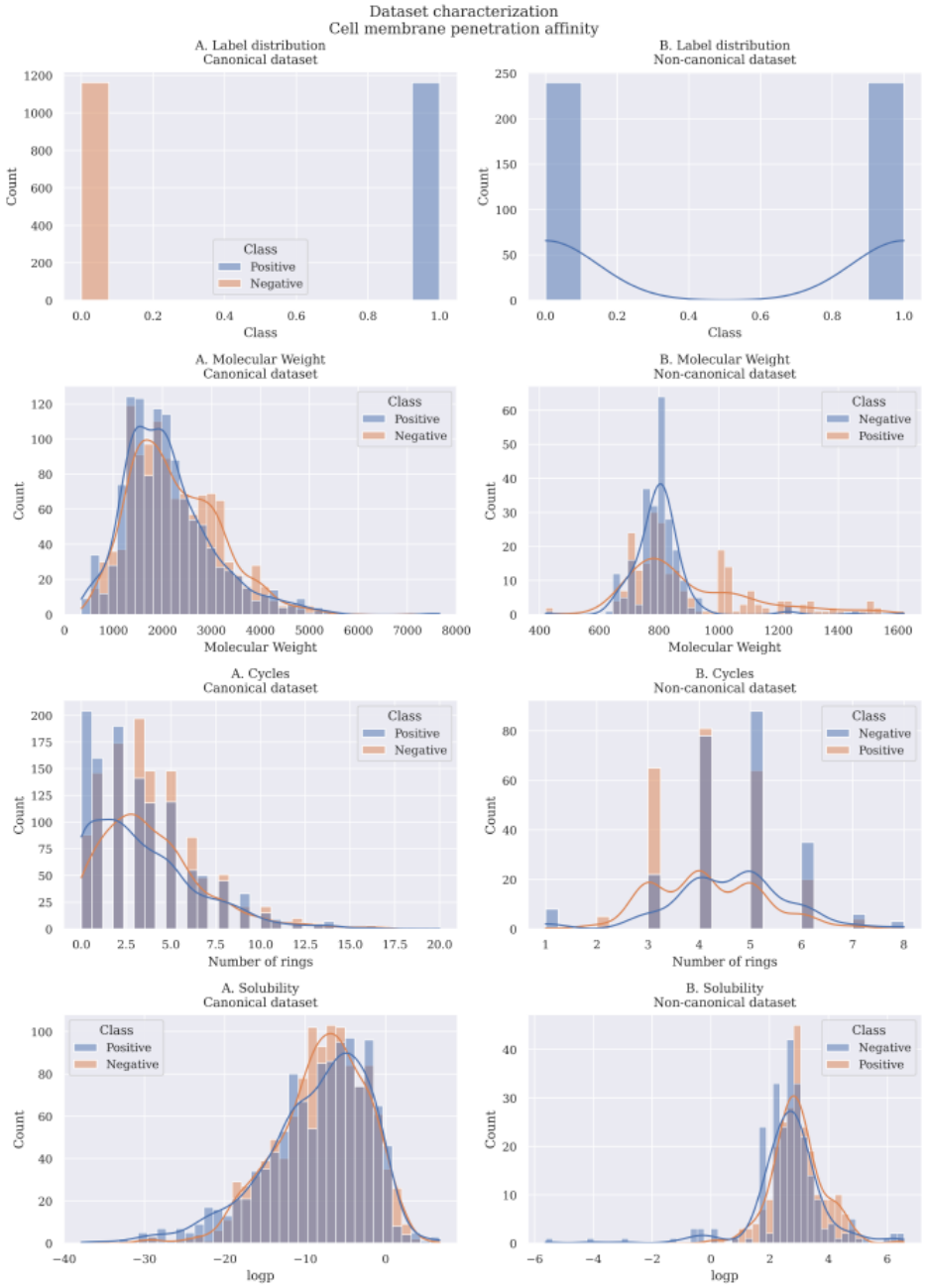


Figure 3: Dataset characterization for the cell penetration datasets.

Training/test splitting

- For each dataset, Hestia-GOOD is used to select the optimal similarity function for dataset partitioning.
- The framework provides quantitative metrics for determining whether a similarity function leads to biochemically relevant partitions.
- We ensured a complete independence of the test sets and allowing for multiple similarity thresholds.

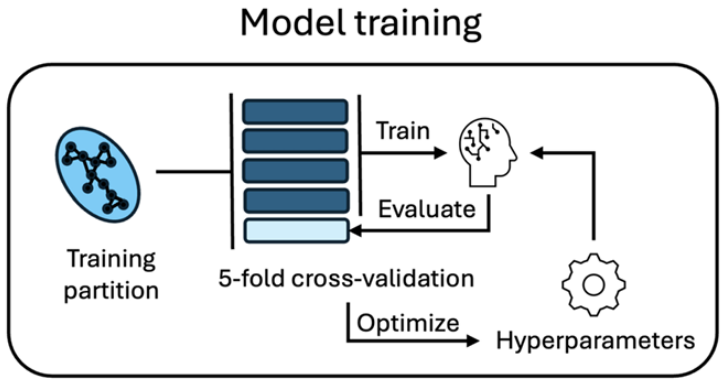


Fig. 2 Visual representation of the model training module

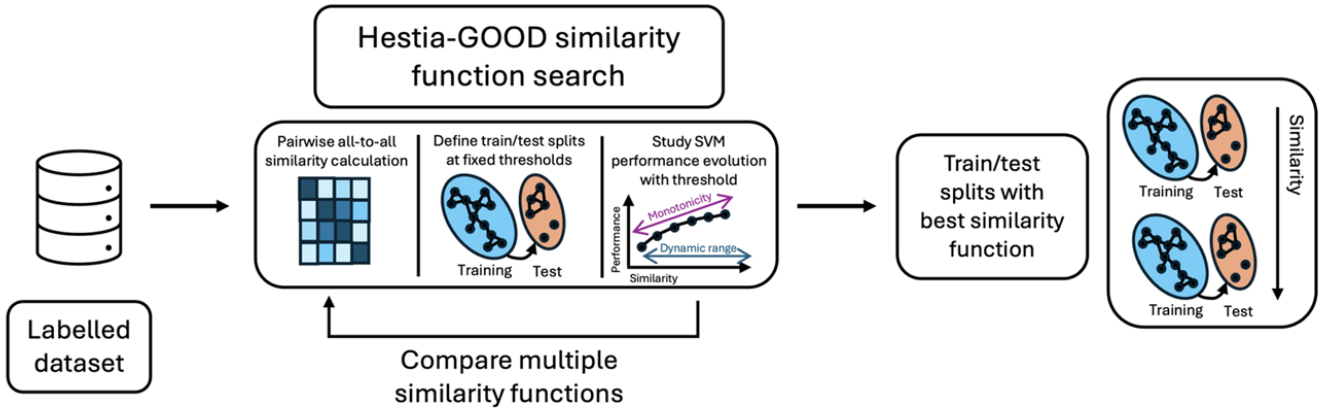


Fig. 3 Identification of the best similarity function for peptide dataset partitioning (1–6) and evaluation of representation methods (7 and 8) using the Hestia-GOOD framework

Interpolation vs extrapolation

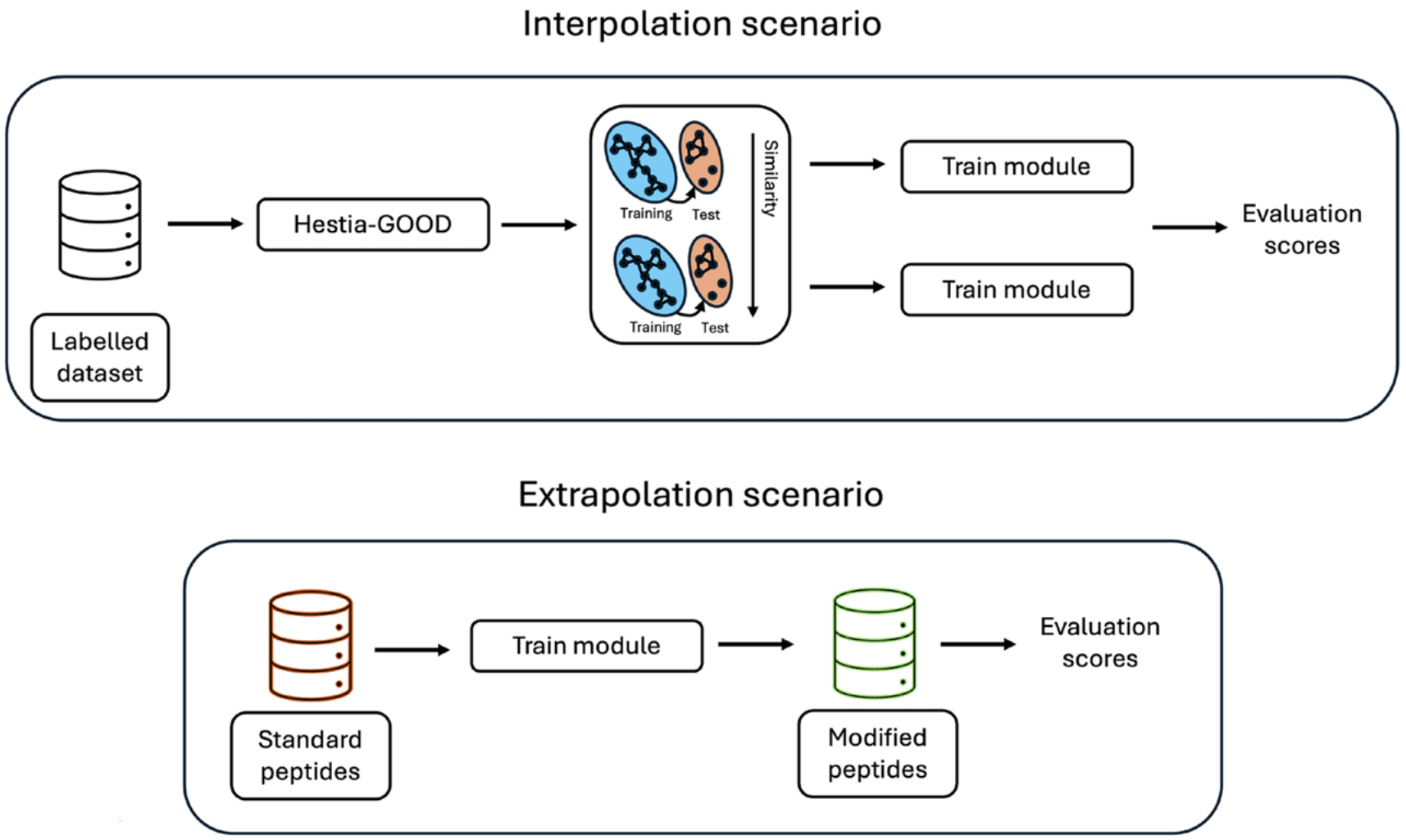
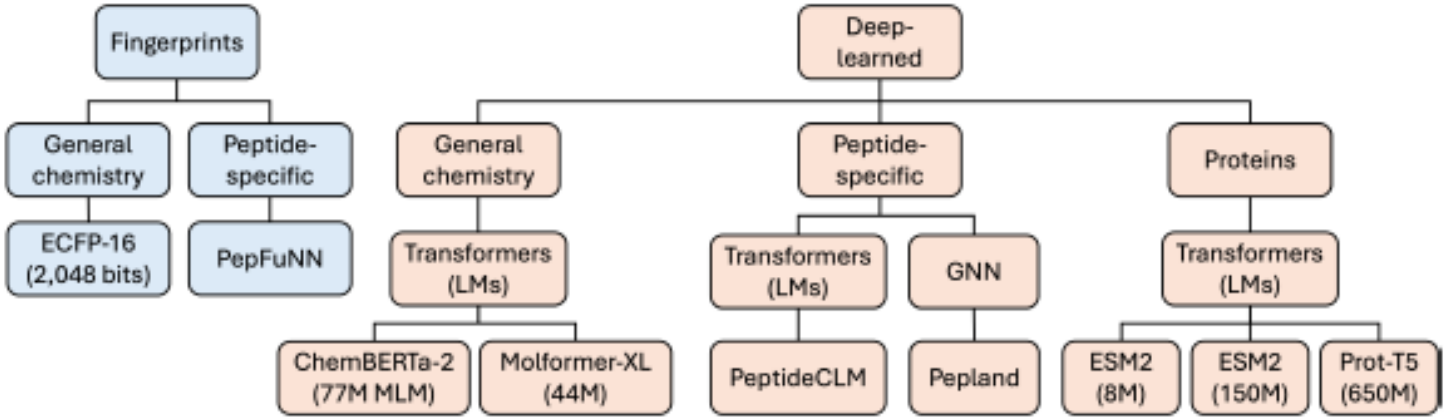


Fig. 4 Difference between the workflows in the interpolation and extrapolation scenarios

Representations

- The resulting partitioned datasets were used for studying which representation performed best across all datasets. We also explored whether the combination of multiple representations could further improve model performance.
- Modern representation techniques rely mostly on self-supervised pre-trained deep learning models, divided by architecture into transformer-based or GNN-based models, and by their pre-training data into general chemistry, peptide-specific, or protein-specific models.



Similarity results: MAPc is an ideal fingerprint

Molecular similarity functions for dataset partitioning are better suited for both canonical and non-canonical peptide datasets than sequence alignments.

Table 2 Best similarity function identified for each dataset considered. ↑: Greater is better. [a] Difference between the minimum and maximum similarity threshold at which viable train/test partitions can be generated. [b] Spearman's ρ correlation coefficient between model performance and similarity threshold

Dataset	Peptide type	Task	Similarity Type	Similarity	Dynamic range (↑) [a]	Monotonicity (↑) [b]
Protein-peptide binding affinity	Standard	Regression	Chemical FP	MAPc-8	70 %	0.8 ± 0.1
Protein-peptide binding affinity	Modified	Regression	Chemical FP	MAPc-20	80 %	0.95 ± 0.03
Cell penetration	Standard	Classification	Chemical FP	MAPc-8	60 %	0.98 ± 0.04
Cell penetration	Modified	Classification	Chemical FP	MAPc-12	60 %	0.5 ± 0.2
Antibacterial	Standard	Classification	Chemical FP	MAPc-8	60 %	0.97 ± 0.02
Antibacterial	Modified	Classification	Chemical FP	ECFP-12	50 %	0.9 ± 0.1
Antiviral	Standard	Classification	Sequence Alignment	MMSeqs2	80 %	0.96 ± 0.05
Antiviral	Modified	Classification	Chemical FP	MAPc-12	70 %	0.6 ± 0.2

Interpolation results

- The choice of representation technique has to be informed by the downstream models, with simpler models favoring linear representations like molecular fingerprints, as in the case of SVM and monomer-based fingerprints like PepFuNN or ECFP for both canonical and non-canonical

- Protein language models are better than peptide-focused representation techniques for modeling canonical peptides, and slightly superior to chemical language models.

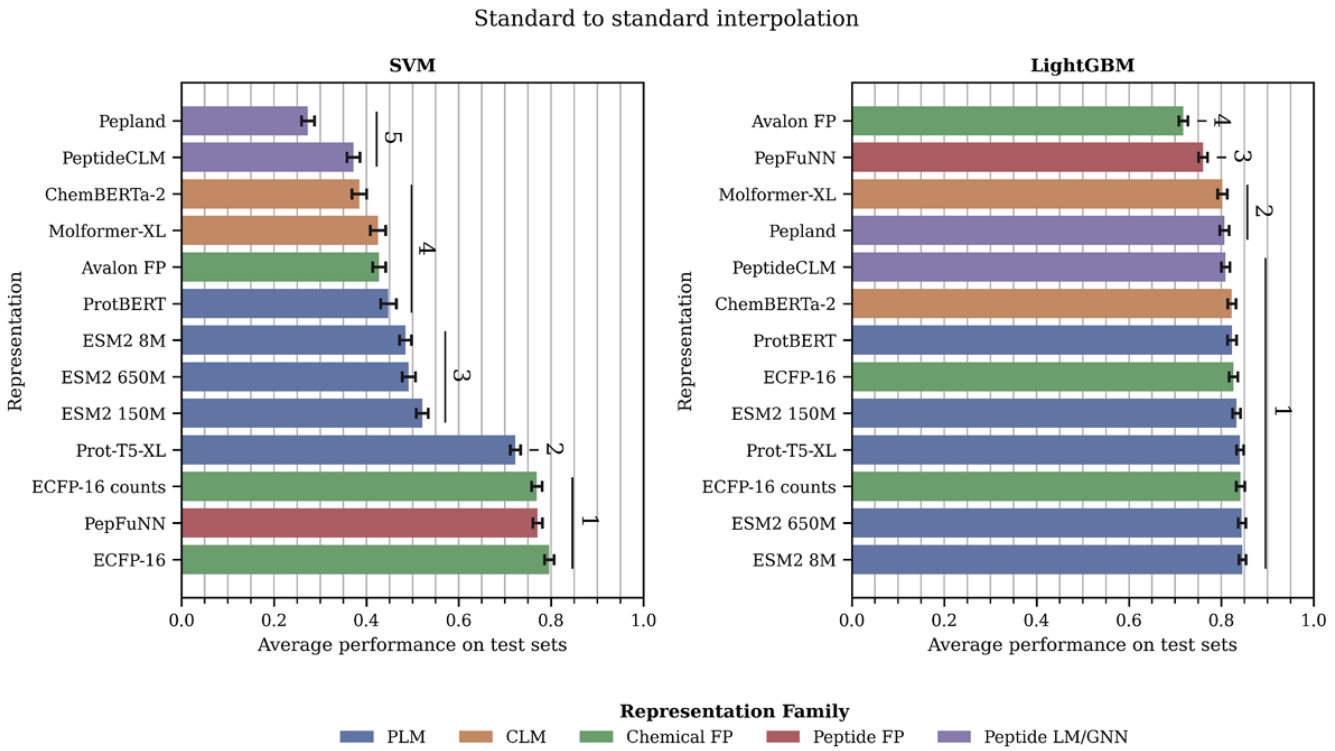


Fig. 6 Evaluation of peptide representations in the standard to standard interpolation. Values are average performance (MCC for classification and SPCC for regression) across all training/testing splits at the different similarity thresholds of each dataset. Error bars: standard deviation. Vertical lines and numbers represent the significant rank each method belongs to. A significant rank is a group of models that are not significantly different ($p > 0.004$) from the best model in the group

Interpolation results

- More complex models like Light Gradient Boosting achieving better overall performance with protein language models for canonicals, and chemical language for non-canonicals (i.e Molformer-XL)

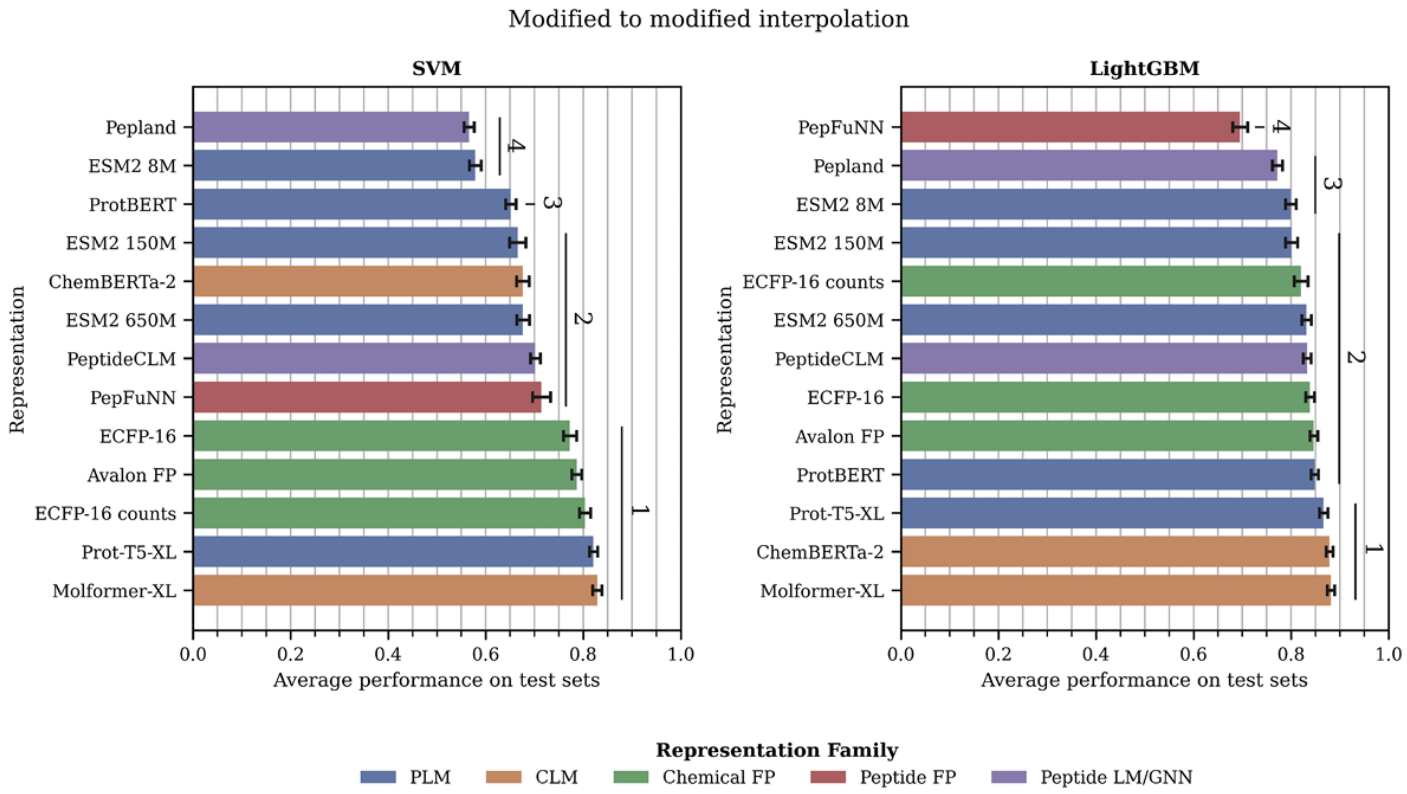


Fig. 7 Evaluation of peptide representations in the modified to modified interpolation. Values are average performance (MCC for classification and SPCC for regression) across all training/testing splits at the different similarity thresholds of each dataset. Error bars: standard deviation. Vertical lines and numbers represent the significant rank each method belongs to. A significant rank is a group of models that are not significantly different ($p > 0.004$) from the best model in the group

Joint modelling - extrapolation

- Finally, we considered whether models exclusively trained on canonical data or non-canonical data could generalise to each other.

- Models trained on canonical datasets are not able to generalize to non-canonical data, and vice versa. However, models trained on joint datasets can generalize to both types of peptides.

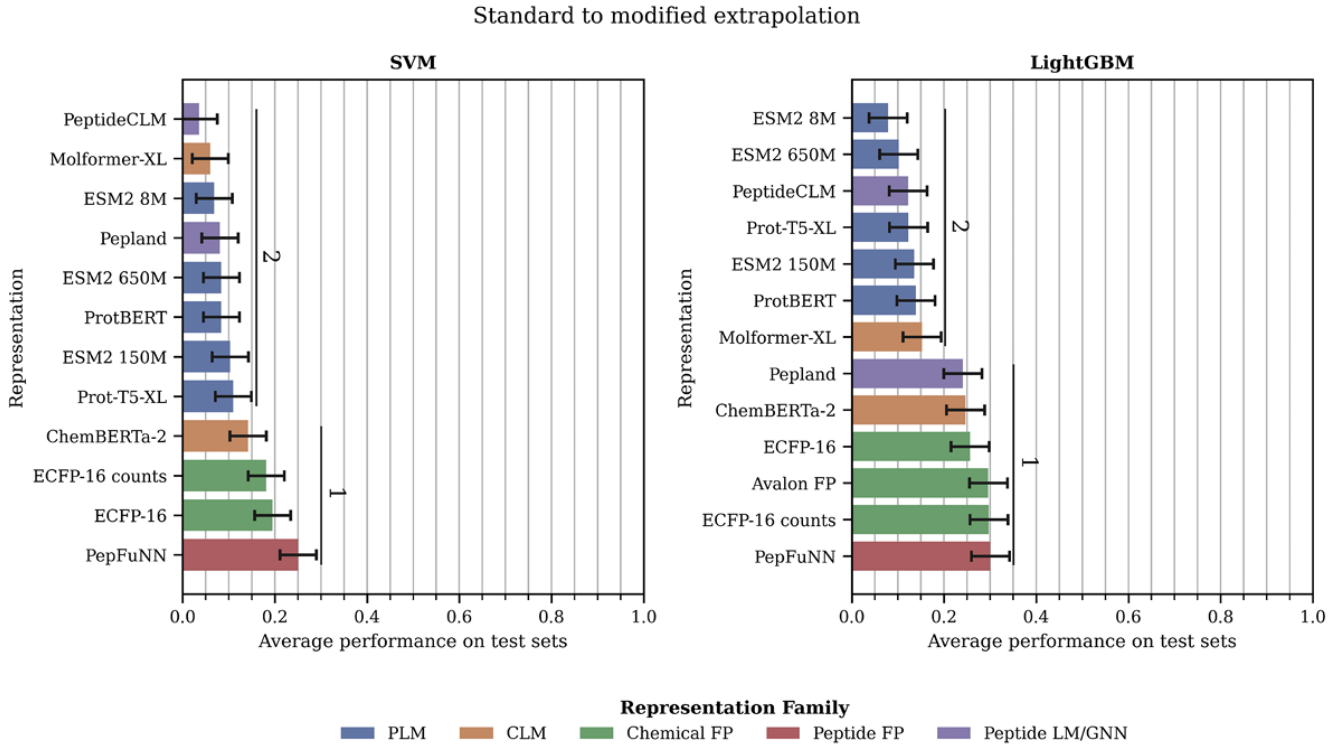


Fig. 8 Evaluation of peptide representations in the standard to modified extrapolation scenario. Values are average performance (MCC for classification and SPCC for regression) across all 25 independent runs. Error bars: standard deviation. Vertical lines and numbers represent the significant rank each method belongs to. A significant rank is a group of models that are not significantly different ($p > 0.05$) from the best model in the group

How to use this in an industrial setup

- We can create foundational models able to interpolate and extrapolate depending on the chemical modifications
- We can generate massive amounts of data to enable AI to design out-of-the-box sequences based on the collected SAR
- In Novo we are using new technologies like IVTT to increase the high-throughput screening to improve the active learning approach and generate more valuable data

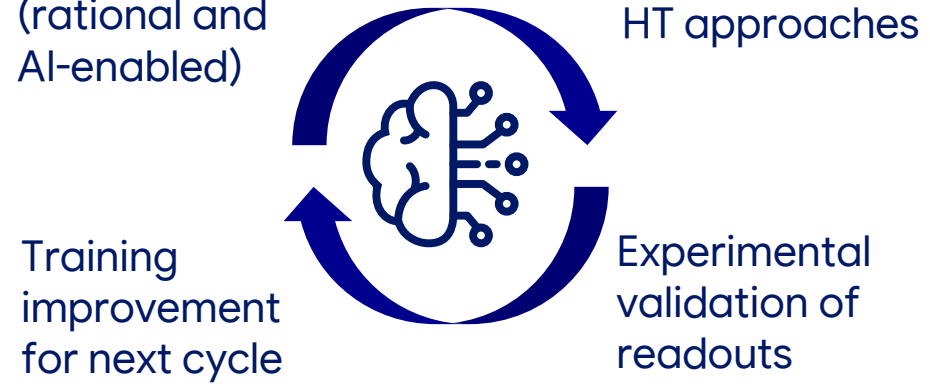
High throughput screening and active learning

IVTT High throughput screening for peptide design

IVTT platform
Fast, flexible, cell-free protein production

- DNA ordering
- IVTT Expression
- Purification
- Cleavage | Desalting

Active learning methodologies based on defined design cycles



Notes go here

PepFuNN: open-source toolkit for peptide analysis

Open Access Article

PepFun: Open Source Protocols for Peptide-Related Computational Analysis

by Rodrigo Ochoa [1] and Pilar Cossio [1,2,*]

[1] Biophysics of Tropical Diseases, Max Planck Tandem Group, University of Antioquia, Medellin 050010, Colombia
[2] Department of Theoretical Biophysics, Max Planck Institute of Biophysics, 60348 Frankfurt am Main, Germany
* Author to whom correspondence should be addressed.

Molecules **2021**, *26*(6), 1664; https://doi.org/10.3390/molecules26061664

Research Article

Pepfun 2.0: Improved Protocols for the Analysis of Natural and Modified Peptides

Rodrigo Ochoa

Article: FDD82 | Received 03 Apr 2023, Accepted 09 Aug 2023, Published online: 24 Aug 2023

RESEARCH ARTICLE | Open Access

PepFuNN: Novo Nordisk Open-Source Toolkit to Enable Peptide in Silico Analysis

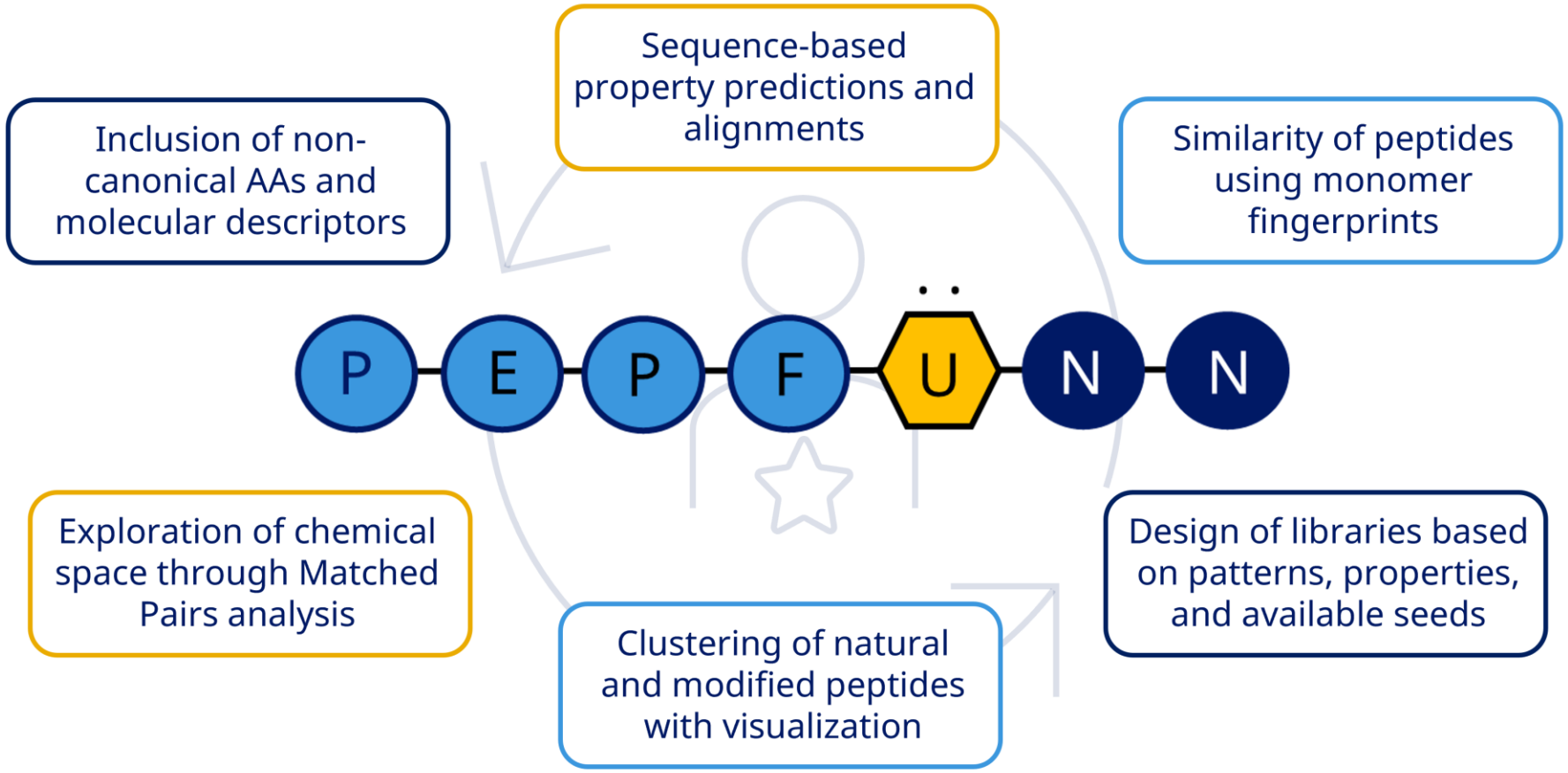
Rodrigo Ochoa, Kristine Deibler

First published: 07 January 2025 | https://doi.org/10.1002/psc.3666

All authors are full time employees at Novo Nordisk A/S.

PepFuNN (Novo version)

- Emphasis in SAR analysis and library design

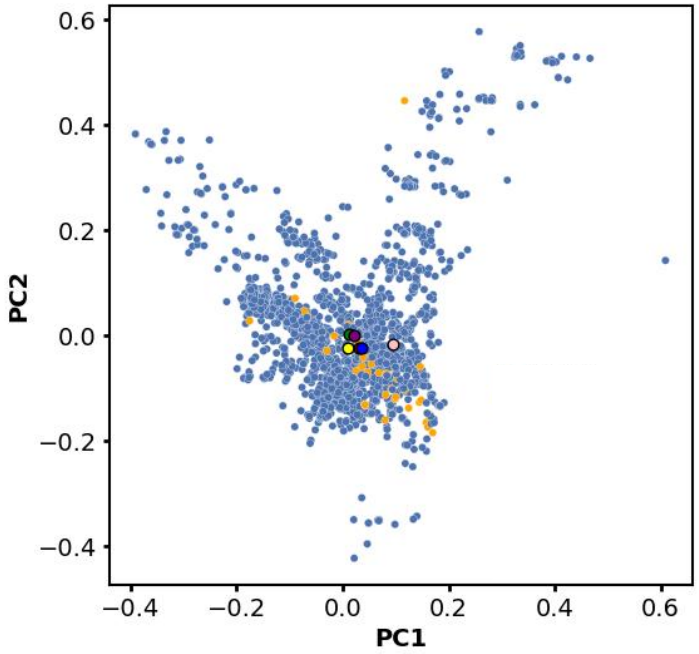


Chemical distances

- Metrics to compare how close the molecules are to each other. These are split into chemical fingerprints, sequence distances (number of mutations), and descriptors Euclidean distances

Strategies:

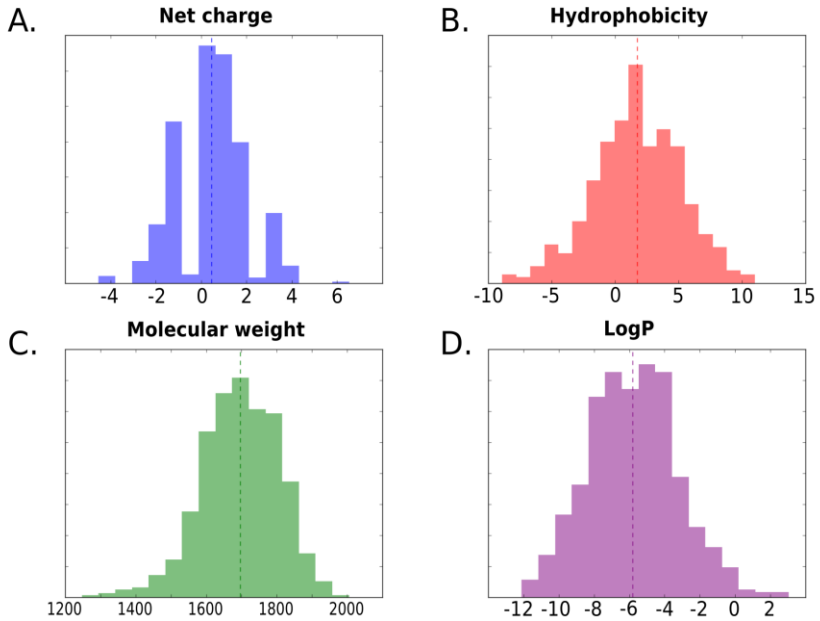
- Use MolFiles to calculate fingerprints (Morgan, MAP4) and compare molecules (RDKit)
- Molecule main chain sequence with natural AAs can be compared using Mafft/Hamming distances (mutations).
- Descriptor Euclidean distance – based on a full set of descriptors (phys-chem properties from RDKit)



Calculation of properties

Table 1 Average values of properties calculated with PepFun for the datasets of peptides associated to the MHC class II protein and the Granzyme B protease.

Property	Average values MHC set	Average values Protease set
Net charge	0.457	-1.872
Molecular weight	1696.111	884.133
LogP	-5.810	-4.235
Hydrophobicity	1.735	0.002
Aromaticity	0.099	0.043
Instability index	33.883	37.095
Isoelectric point	7.290	4.375
Number hydrogen donors	24.183	13.337
Number hydrogen acceptors	23.670	13.578
Solubility rules failed	2.046	1.793
Synthesis rules failed	4.504	0.544



- Use SMILES and RDKit molecular objects to predict physico-chemical properties
- Use of amino acid scales and empirical rules to filter peptide sequences, as well as amino acid scales

Alignments & descriptors

A.

Peptide 1: WWSEVN[1OL]AEF

Peptide 2: KTEEISEVN[STA]VAEF

```
W W - - - - - - S E V N 1OL  -  - A E F
                  | | | |          | | |
- - K T E E I S E V N  -  STA V A E F
```

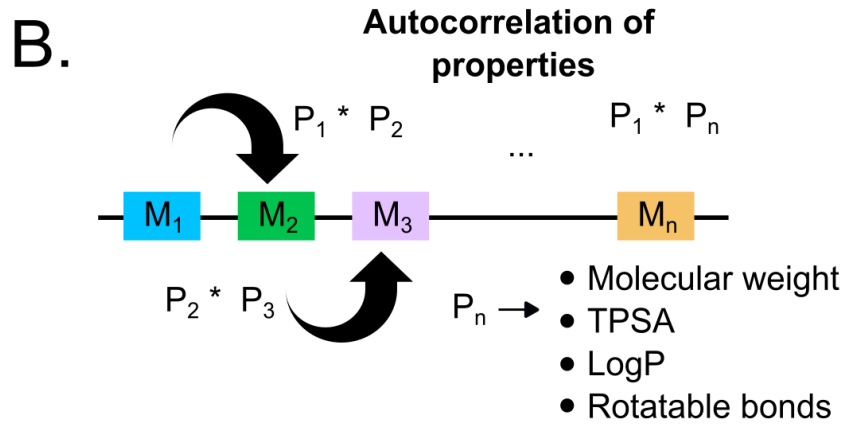
Alignment with or without a scoring matrix
(Based on the HELM monomer dataset)

Biopython pairwise alignment was adapted to include NCAAs and a similarity-based scoring matrix

Amino acid-based descriptors can be calculated based on physico-chemical properties of the residues (include also NCAAs)

This can be used to create explainable models to design peptides under certain hypothesis

B.



Library design based on requirements

- 2D-based strategy developed to accelerate the screening of peptides under certain chemical considerations, and split in 3 main phases

Seed exploration
(initial candidates based on some conditions)

Scanning
(combinatorial replacements following a path)

Refinement
(more oriented modifications)

+

Optimization algorithm

(Using multiple fitness options)

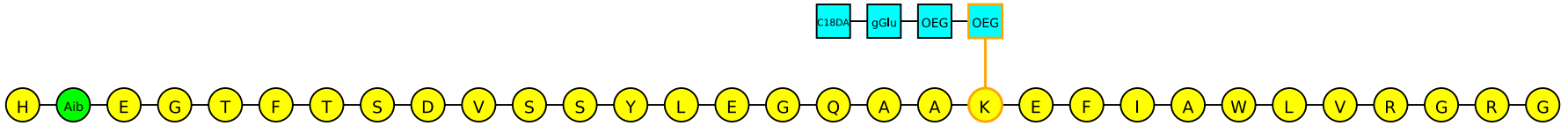
- Experimental results
- ML model
- Structure-based calculations

Representation: BI Line Notation (BILN)

- An alterative to notations like HELM, but more human (MedChem) readable to allow all kinds of text-based searches, comparisons and analyses

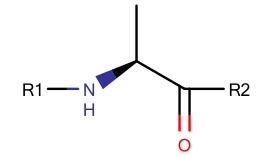
Example (semaglutide):

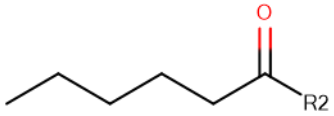
H-Aib-E-G-T-F-T-S-D-V-S-S-Y-L-E-G-Q-A-A-K(1,3)-E-F-I-A-W-L-V-R-G-R-G.C18DA-gGlu-OEG-OEG(1,2)



Goals:

- **Unambiguous, extensible, human-readable**
- Needs a monomer library (e.g. HELM monomers):





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BILN: A Human-Readable Line Notation for Complex Peptides

Thomas Fox*, Michael Bieler, Peter Haebel, Rodrigo Ochoa, Stefan Peters, and Alexander Weber

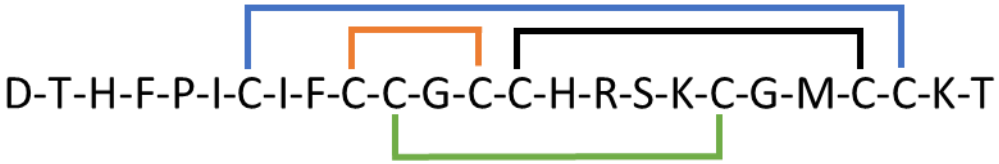
Characteristics and examples

- SMILES-like representation, where the monomers are the basic building blocks
- We assume a N->C terminal direction of AA for easy readability
- "-" bond between monomers (from R2 in monomer *n* to R1 in monomer *n+1*)
- Any additional bonds (branching, cycles) within parenthesis after the monomer

Example	BILN
Simple linear peptide	P-E-P-T-I-D-E
Peptide with internal C-C disulfide bridge and amidation at the C-terminus	A-C(1,3)-G-A-G-C(1,3)-D-NH2
C-term to N-term cyclic peptide	C(1,1)-Y-C-L-I-C(1,2)
Same peptide with C-C disulfide bridge	C(1,3)-Y-C-L-I-C(1,3)
Branched peptide	A-G-K(1,3)-D-A.C18DA-G-G(1,2)

A more complex example

Hepcidin: an antimicrobial peptide expressed in the liver, 25 amino acid long whose structure is stabilized by four disulfide bridges



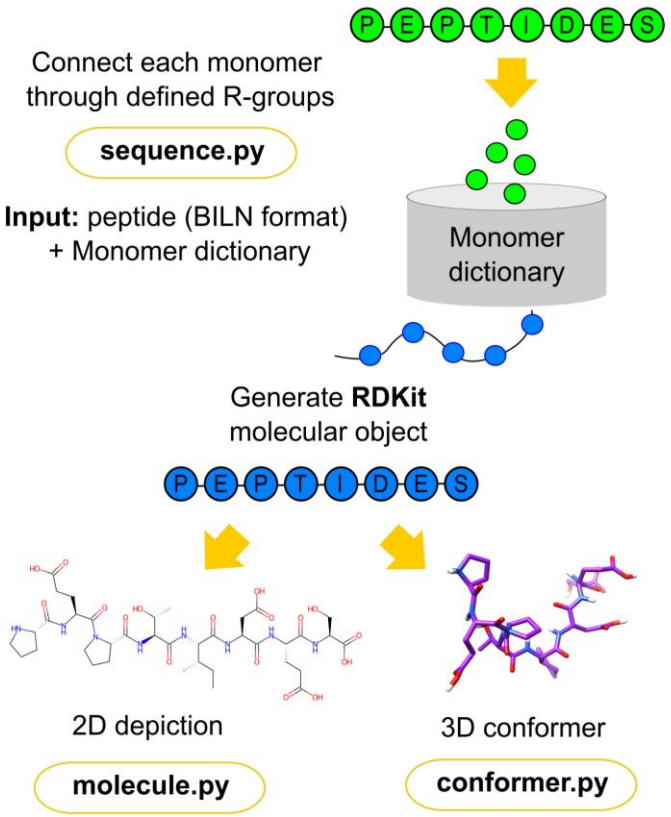
PEPTIDE1{D.T.H.F.P.I.C.I.F.C.C.G.C.C.H.R.S.K.C.G.M.C.C.K.T}$PEPTIDE1,PEPTIDE1,7:R3-23:R3|PEPTIDE1,PEPTIDE1,10:R3-13:R3|PEPTIDE1,PEPTIDE1,11:R3-19:R3|PEPTIDE1,PEPTIDE1,14:R3-22:R3$$$

D-T-H-F-P-I-C(1,3)-I-F-C(2,3)-C(3,3)-G-C(2,3)-**C(4,3)**-H-R-S-K-C(3,3)-G-M-**C(4,3)**-C(1,3)-K-T

- How can I analyze this format as a molecular object?

pyPept

- A python package to allow the analysis of natural and modified peptides that are assembled based on personalized monomer dictionaries using BILN (also HELM or FASTA)

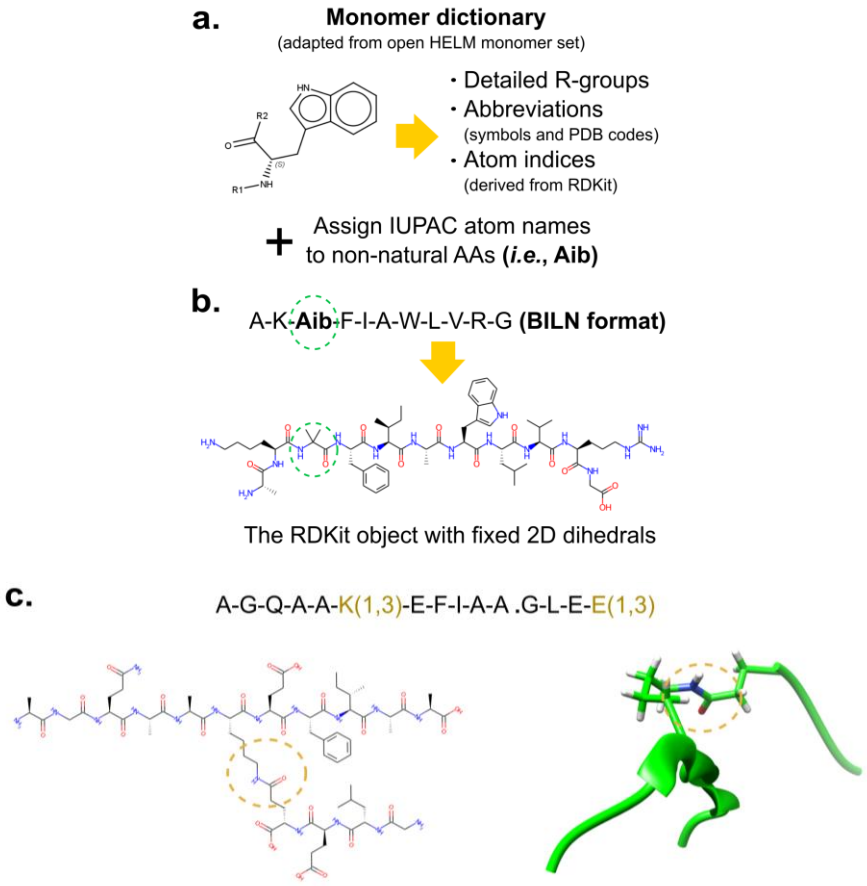


Software | Open access | Published: 12 September 2023

pyPept: a python library to generate atomistic 2D and 3D representations of peptides

Rodrigo Ochoa, J. B. Brown & Thomas Fox

Journal of Cheminformatics **15**, Article number: 79 (2023) | Cite this article



AI-driven structure prediction for cyclic peptides

RSC Medicinal Chemistry

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RESEARCH ARTICLE

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Cite this: DOI: 10.1039/d6md00676k

Benchmarking AI-driven protocols for cyclic peptide conformer prediction and ranking

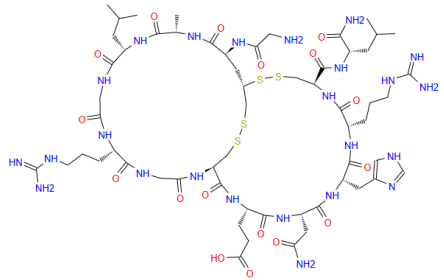
Jovan Damjanovic *[a] and Rodrigo Ochoa *[b]

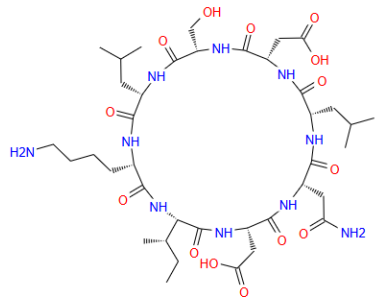
Datasets

- Integrase bound to cyclic peptides with known structures and head-to-tail topologies.
- Compstatin peptide bound to C3c receptor, and variants without structure but affinity data available
- Hydrolase target bound to cyclic peptides with disulfide bonds. Additional variants with affinity available.
- Cyclic peptides in solution with experimental structures available in public GitHub repo
- **Some peptides contain non-natural amino acids, and some targets have more than one chain.**

Table 3 Selected systems for head-to-tail cyclic peptides bound to HIV integrase. All the complexes report structures in the PDB, but due to the presence of head-to-tail topologies, both the sequence and the SMILES were used as input during the benchmark[20]

PDB code/internal code	Sequence	IC_{50} (uM)	Publication (DOI)
3av9 (seq and SMILES)	SAKIDNLD	435	https://doi.org/10.1002/cbic.201100350
3ava (seq and SMILES)	ALKIDNLD	39.7	
3avb (seq and SMILES)	SLKIDNLD	85.2	
3avf (seq and SMILES)	DLKIDNLD	295	
3avg (seq and SMILES)	ADKIDNLD	>500	
3avh (seq and SMILES)	ARKIDNLD	>500	
3avi (seq and SMILES)	SLKIDNMD	>500	
3avj (seq and SMILES)	ALKIDNMD	>500	
3avk (seq and SMILES)	SLKIDNED	>500	
3avm (seq and SMILES)	SRKIDNLD	>500	
3avn (seq and SMILES)	SHKIDNLD	>500	
3avl (seq and SMILES)	ATKIDNLD	65.8	
3wne (seq and SMILES)	PKIDNG	>10 000	
3zgc (seq and SMILES)	GDEETGE	0.479	
5xn3 (seq and SMILES)	RGDINNNV	0.671	





Extra methods to convert all-atom output to AA-like format, and viceversa

Structure prediction tools

- HighFold[1], AlphaFold2 like in EvoBind2[2], CycPepDock[3], Boltz-1 and 2[4], and Chai-1[5]

[1] *Zhang, Chenhao, et al. HighFold: accurately predicting structures of cyclic peptides and complexes with head-to-tail and disulfide bridge constraints. Briefings in Bioinformatics,* ***2024****, 25.3: bbae215.*
[2] *Li, Qiuzhen, Efstathios Nikolaos Vlachos, and Patrick Bryant. Design of linear and cyclic peptide binders of different lengths only from a protein target sequence. bioRxiv,* ***2024****: 2024-06.*
[3] *Kosugi, Takatsugu, and Masahito Ohue. Design of Cyclic Peptides Targeting Protein–Protein Interactions Using AlphaFold. International Journal of Molecular Sciences,* ***2023****, 24.17: 13257.*
[4] *Wohlwend, Jeremy, et al. Boltz-1: Democratizing Biomolecular Interaction Modeling. bioRxiv, 2024:* ***2024****-11.*
[5] *Chai Discovery, et al. Chai-1: Decoding the molecular interactions of life. bioRxiv,* ***2024****: 2024-10.*

- **Considerations:**
  - Boltz-1 and 2, and Chai-1 can accept SMILES as input for modified peptides
  - For methods without multimer capabilities, an artificial linker was used between the targets (e.g., CycPepDock)
  - Some tools were run with an optimized configuration different from the default setup after initial attempts

In the monomer regime, dedicated cyclic peptide tools perform the best

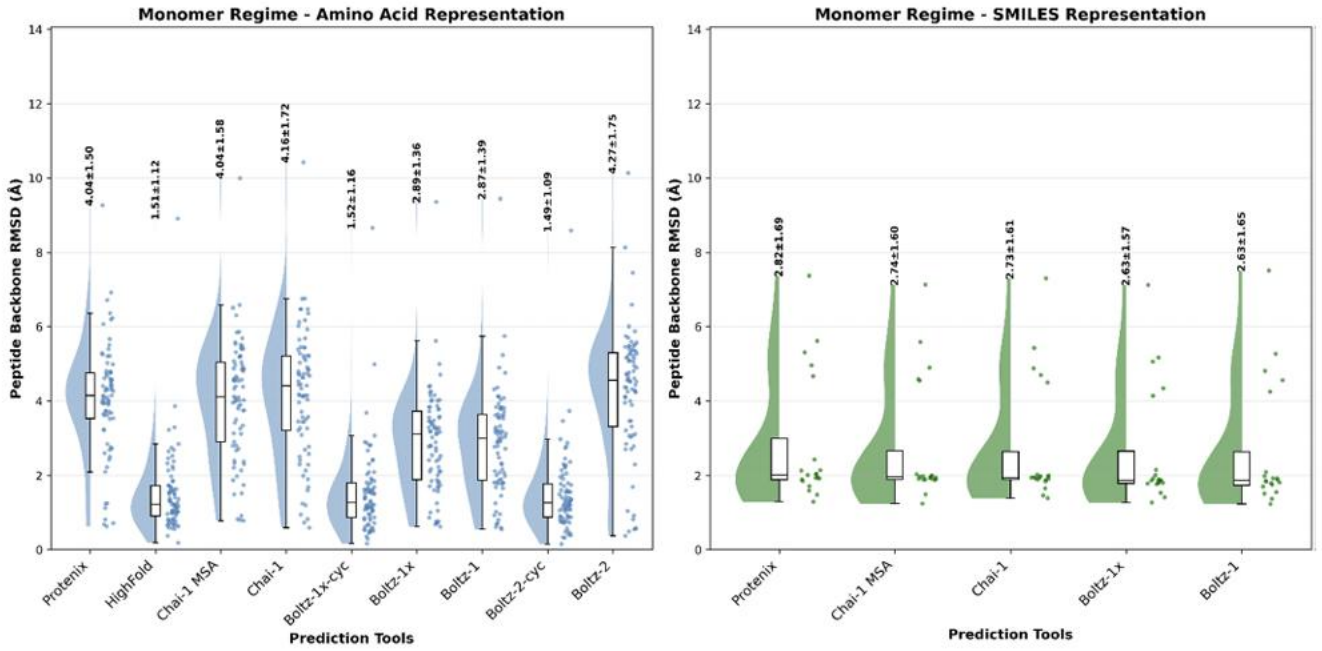


Fig. 1 Structural similarity estimated by average RMSD (Å) of predicted cyclic peptide structures to the reference structures in the monomer regime (no target protein). Only tools with monomer capability (both panels) and only tools with SMILES input capability (right hand side) were considered.

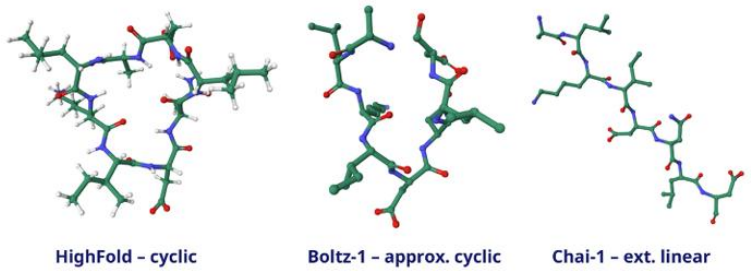


Fig. 2 Comparison of representative cyclic peptide structure predictions (reference: PDB 3ava) provided by different tool families. HighFold, given its relative positional encoding offset, cyclizes the peptide correctly. Boltz-1 seemingly "remembers" the bound conformation and approximately cyclizes the peptide in it despite no target protein present. Chai-1 produces a completely extended linear peptide.

- Prediction tools incorporating a relative positional encoding tweak perform better
- This is an intuitive result – without constraints or a binding partner, one may expect an unfolded linear peptide
  - Chai1 returns exactly that, whereas Boltz-1 seems to reproduce lower end-to-end distances, potentially due to memorization

In the multimer regime, the AA representation outperforms SMILES and the bound peptide conformation is captured more easily

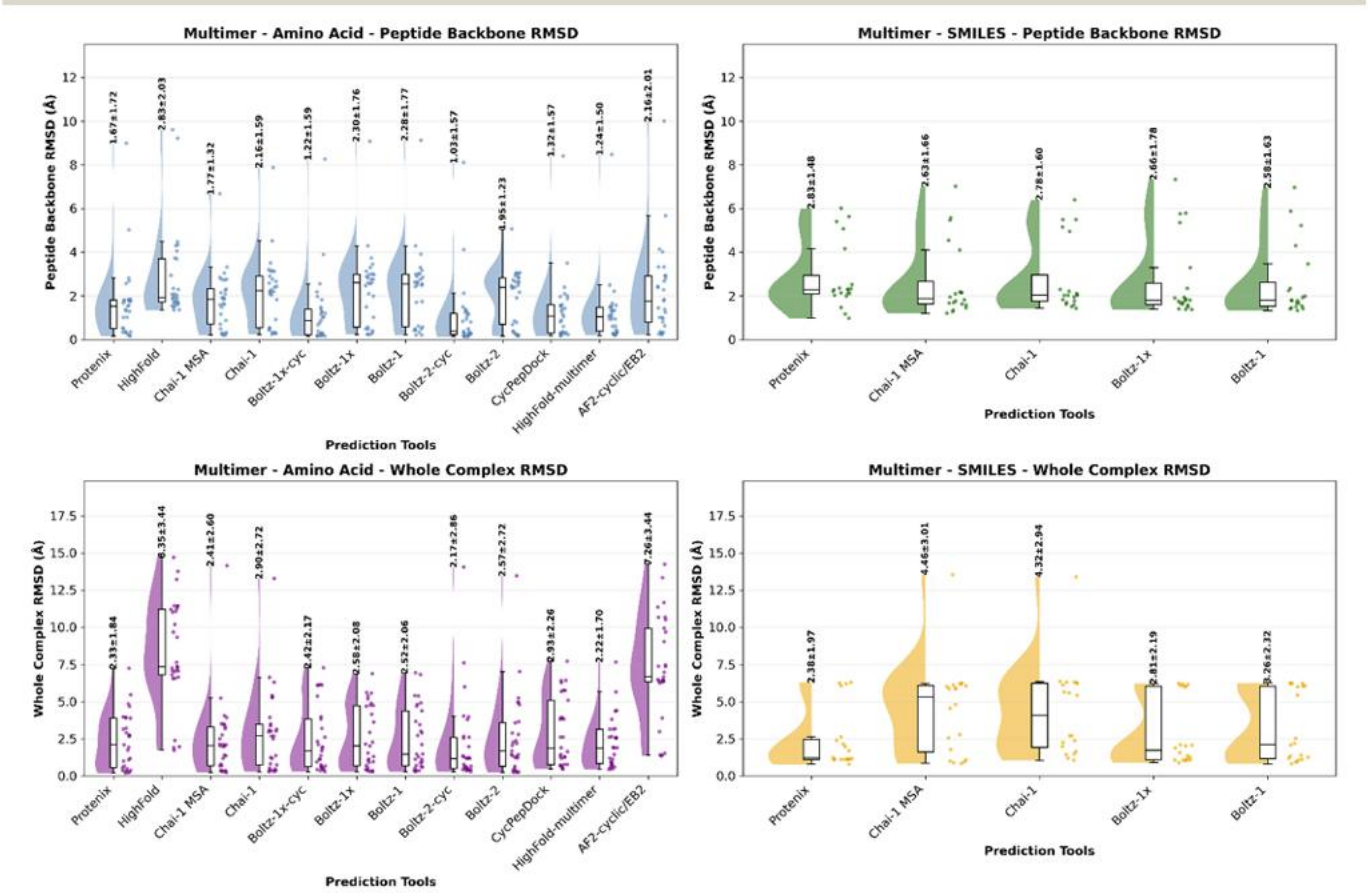


Fig. 3 Performance comparison of cyclic peptide structure prediction tools in the multimer regime (target protein included) based on average RMSD (Å). Only peptides with known binding partners and bound experimental structures were included. The same inclusion constraints apply as in Fig. 1.

- In the presence of the protein target, the relative positional encoding tweak is not required for the peptide to be (at least approximately) cyclized
- The AA representation tends to outperform SMILES
- CycPepDock displays a strong dependence on parameters, with the default settings (single-seq MSAs, AF2M_v2 weights) performing quite poorly, and adjusted settings doing reasonably well

Prediction quality may not correlate with experimental binding affinity

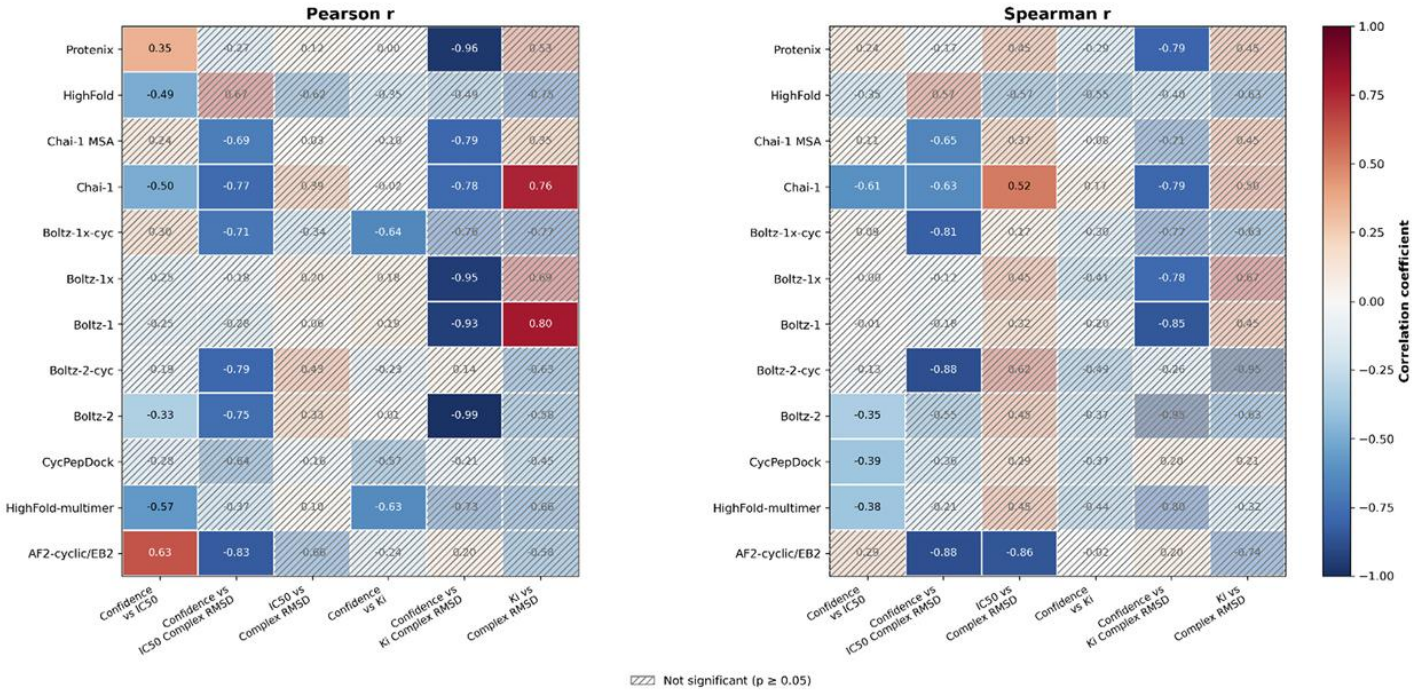


Fig. 5 Complex pLDDT and structure prediction accuracy comparison to experimentally determined potency (IC_{50}, where applicable) or binding affinity (K_i, where applicable), with outliers (data reported outside assay sensitivity limit) excluded. Results with outliers included are shown in Fig S2, and correlation metrics for the two sets of results are shown in Tables S4 and S5, respectively. pTM and ipTM show similar patterns to pLDDT (data not shown).

- One may intuitively assume that the better the binder a peptide is, the more likely the complex structure is to be predicted accurately
- With the caveat that the sample size is small, this does not appear to be the case on the HIV integrase data set
- The trend seen for RMSD also holds true for pLDDT and other model confidence metrics, suggesting model confidence might also not correlate with binding affinity

Conclusions

- **Match models to peptide chemistry.** Representation choice and training-data coverage determine predictive performance.
- **Evaluate generalization rigorously.** Similarity-aware splitting helps distinguish interpolation from extrapolation.
- **Connect AI design with experiments.** Active learning can reduce testing effort and shorten design cycles.
- **Use structure predictions cautiously.** Performance depends on topology and binding context. Confidence alone does not establish affinity.

Thanks

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